

Data Compression using Singular Value Decomposition (SVD)

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1 Singular Value Decomposition

1.1 Introduction

Singular Value Decomposition (SVD) is a matrix factorization technique that decomposes a matrix into three matrices. For a matrix $A \in \mathbb{R}^{m \times n}$, its singular value decomposition is

$$A = U \Sigma V^T,$$

where

$$U \in \mathbb{R}^{m \times m}, \quad \Sigma \in \mathbb{R}^{m \times n}, \quad V \in \mathbb{R}^{n \times n}.$$

The matrices U and V are orthogonal matrices, satisfying

$$U^T U = I_m, \quad V^T V = I_n,$$

while Σ is a diagonal matrix containing the singular values of A .

The singular values are conventionally arranged in non-increasing order:

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r \geq 0,$$

where

$$r = \text{rank}(A).$$

Equivalently, the decomposition can be written as

$$A = \sum_{i=1}^r \sigma_i u_i v_i^T,$$

where u_i and v_i are the corresponding left and right singular vectors.

1.2 Connection to Eigenvalue Decomposition

The singular values of A are related to the eigenvalues of $A^T A$.

Since

$$A = U \Sigma V^T,$$

we obtain

$$A^T A = V \Sigma^T U^T U \Sigma V^T.$$

Using the orthogonality of U ,

$$U^T U = I,$$

giving

$$A^T A = V \Sigma^T \Sigma V^T.$$

Therefore, V contains the eigenvectors of $A^T A$, while the eigenvalues are the squared singular values:

$$\lambda_i = \sigma_i^2.$$

Consequently,

$$\sigma_i = \sqrt{\lambda_i}.$$

This relationship provides one way of computing the SVD and forms the basis for the eigenvalue-based approach explored in this project.

2 Step 1: Householder Reflector Matrix

In the eigenvalue-based computation of the SVD, the symmetric matrix

$$B = A^T A$$

is first constructed.

Geometrically, a Householder reflection matrix reflects a vector across a hyperplane perpendicular to a given vector.

For a nonzero vector u , a Householder reflector is generally defined as

$$H = I - 2 \frac{uu^T}{u^T u}.$$

If u is a unit vector, then $u^T u = 1$, and the expression simplifies to

$$H = I - 2uu^T.$$

Every Householder reflector has the following properties:

- Symmetric:

$$H^T = H$$

- Orthogonal:

$$H^T H = I$$

3 Step 2: Tridiagonalization

A symmetric matrix B can be transformed into a symmetric tridiagonal matrix by applying a series of Householder transformations.

Each transformation is an orthogonal similarity transformation of the form

$$B_{k+1} = H_k B_k H_k^T,$$

where H_k is a Householder reflector.

After applying the required sequence of transformations, the matrix is reduced to a symmetric tridiagonal matrix of the form

$$T = \begin{bmatrix} b_{11} & b_{12} & 0 & 0 & 0 & \dots \\ b_{12} & b_{22} & b_{23} & 0 & 0 & \dots \\ 0 & b_{23} & b_{33} & b_{34} & 0 & \dots \\ 0 & 0 & b_{34} & b_{44} & b_{45} & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}.$$

Since each Householder transformation is orthogonal, the transformation preserves the eigenvalues of the matrix. Therefore,

$$\lambda_i(T) = \lambda_i(B).$$

4 Step 3: QR Iteration

The tridiagonal matrix T obtained from the previous step is used to compute its eigenvalues. This is done using the QR iteration algorithm.

Let

$$T_0 = T.$$

At each iteration, the current matrix T_k is factorized using the QR decomposition:

$$T_k = Q_k R_k,$$

where Q_k is orthogonal and R_k is upper triangular.

The next iterate is then obtained by reversing the order of the factors:

$$T_{k+1} = R_k Q_k.$$

Since

$$T_k = Q_k R_k,$$

we have

$$T_{k+1} = R_k Q_k = Q_k^T T_k Q_k.$$

Therefore, T_{k+1} is orthogonally similar to T_k and has the same eigenvalues.

Repeating this process,

$$T_0 \rightarrow T_1 \rightarrow T_2 \rightarrow \cdots \rightarrow T_k,$$

causes the off-diagonal elements to converge towards zero. Thus,

$$T_k \rightarrow D,$$

where D is a diagonal matrix:

$$D = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix}.$$

The diagonal entries of D are therefore the eigenvalues of the original matrix $B = A^T A$.

The singular values of A can then be obtained from these eigenvalues:

$$\sigma_i = \sqrt{\lambda_i}.$$

5 Step 3: Construct the Eigenvectors of B using the Eigenvalues